

Entity Framework Core 8. 0

Lab 1: Understanding ORM with a Retail Inventory System

Objectives :-

Understand what ORM is and how EF Core helps bridge the gap between C# objects and relational tables.

1. What is ORM?

ORM (Object-Relational Mapping) is a technique that lets you interact with a database using C# classes instead of writing SQL queries directly.

Benefits:

- Productivity: Write less code and focus on business logic, not SQL.
- Maintainability: Changes in your C# code are easier to manage than raw SQL scripts.
- Abstraction: You don't need to know SQL to work with the database.

2. EF Core vs EF Framework :

EF Core

- Cross-platform: Works on Windows, Linux, and macOS.
- Lightweight: Designed for performance and efficiency.
- Modern features: Supports LINQ, async/await, compiled queries, bulk operations, and JSON columns.
- Flexible: Works with many databases (SQL Server, SQLite, PostgreSQL, MySQL, and more).
- Actively developed: Receives frequent updates and new features.
- Best for: New projects, cloud apps, microservices, and cross-platform solutions.

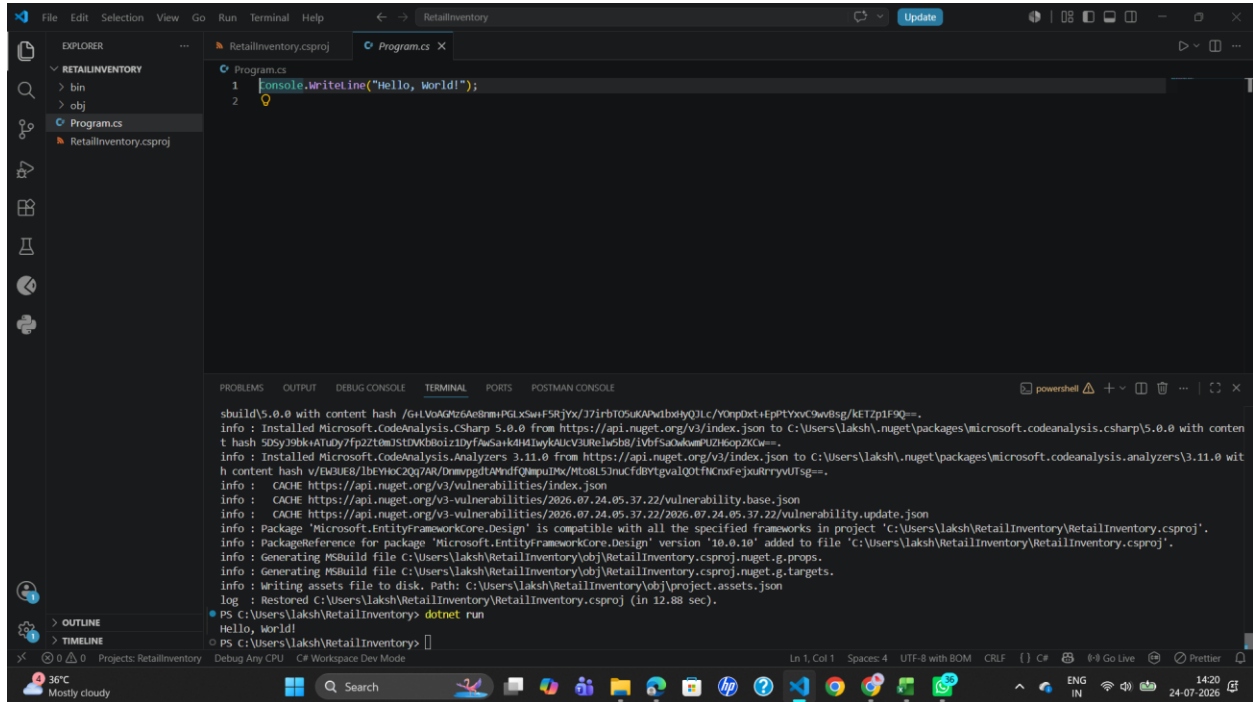
EF Framework (EF6)

- Windows-only: Runs only on the .NET Framework.
- Heavier: More monolithic and less optimized for modern workloads.
- Fewer features: Limited async support, no compiled queries, fewer extensibility points.
- Less flexible: Mainly supports traditional SQL databases.
- Mature and stable: Well-tested and reliable for legacy applications.
- Visual designer: Includes EDMX for database modeling (not in EF Core).
- Best for: Maintaining existing Windows applications or when EDMX is required.

3. EF Core 8.0 Features:

- JSON column mapping: Store and query JSON data directly in SQL Server columns.
- Compiled models: Faster startup and query performance.
- Interceptors: Hook into database operations for logging or custom logic.
- Bulk operations: Improved support for inserting/updating many records at once.

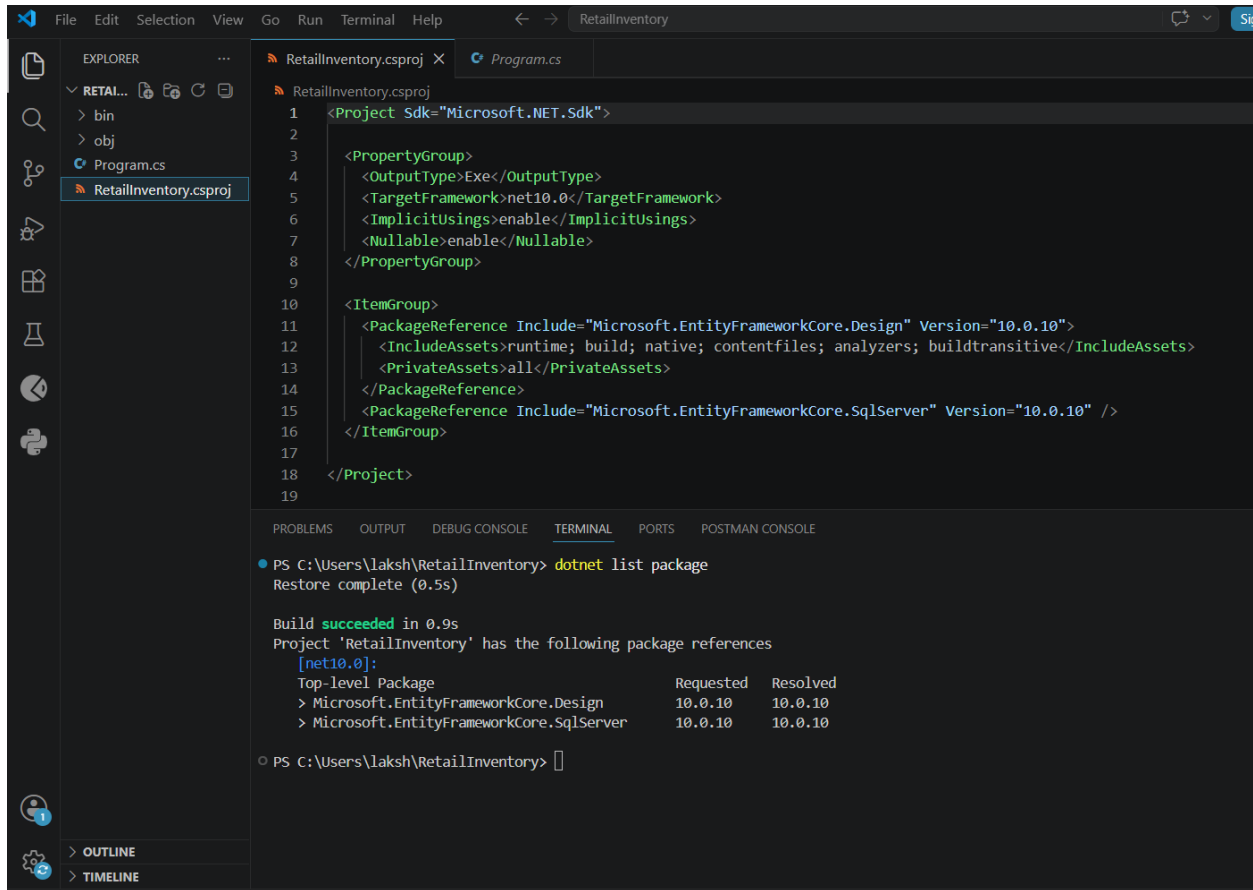
4. Create a .NET Console App :



```
File Edit Selection View Go Run Terminal Help
RetailInventory
Update
EXPLORER
RETAILINVENTORY
  bin
  obj
  Program.cs
  RetailInventory.csproj
Program.cs
1 Console.WriteLine("Hello, world!");
2
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
shuild5.0.0 with content hash /GtLVAGz6Ae8nmPGLxSefF8jYx/771rbT05uKAPv1b8yQJLC/Y0pDxt+EpTYvC0wB8p/kETZp1F9Q==.
Info : Installed Microsoft.CodeAnalysis.CSharp 5.0.0 from https://api.nuget.org/v3/index.json to C:\Users\laksh\.nuget\packages\microsoft.codeanalysis.csharp\5.0.0 with conten
t hash 505y39bki+AtUdy7fp2Zt0mJSTDMKboiz1DyfA6sA+k4H41wyKkUCV3URE1w5b8/1VbfSa0w0mF0Z8opZK0w==.
Info : Installed Microsoft.CodeAnalysis.Analyzers 3.11.0 from https://api.nuget.org/v3/index.json to C:\Users\laksh\.nuget\packages\microsoft.codeanalysis.analyzers\3.11.0 wit
h content hash v/ENQUE8/1bEYh0C2Qq7AR/DnmvpgdUAmdFQmpu1Dk/Mto8L5JnucfdBYtgvalQ0tfnCn0fejx0riryvUTsg==.
Info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
Info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05:37:22/vulnerability.base.json
Info : CACHE https://api.nuget.org/v3-vulnerabilities/2026.07.24.05:37:22/2026.07.24.05:37:22/vulnerability.update.json
Info : Package 'Microsoft.EntityFrameworkCore.Design' is compatible with all the specified frameworks in project 'C:\Users\laksh\RetailInventory\RetailInventory.csproj'.
Info : PackageReference for package 'Microsoft.EntityFrameworkCore.Design' version '10.0.10' added to file 'C:\Users\laksh\RetailInventory\RetailInventory.csproj'.
Info : Generating MSBuild file C:\Users\laksh\RetailInventory\obj\RetailInventory.csproj.nuget.g.props.
Info : Generating MSBuild file C:\Users\laksh\RetailInventory\obj\RetailInventory.csproj.nuget.g.targets.
Info : Writing assets file to disk. Path: C:\Users\laksh\RetailInventory\obj\project.assets.json
log : Restored C:\Users\laksh\RetailInventory\RetailInventory.csproj (in 12.88 sec).
PS C:\Users\laksh\RetailInventory> dotnet run
Hello, world!
PS C:\Users\laksh\RetailInventory>
```

5. Install EF Core Packages:

→ dotnet add package Microsoft.EntityFrameworkCore.SqlServer



→ dotnet add package Microsoft.EntityFrameworkCore.Design